

# Riesz Intertwining and the Two-Term Channel Transport Budget

## Exact Identities Behind the RH-202 Obstruction

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### Abstract

RH-202 showed that the physical edge quartet is not carried between adjacent levels by the unmodified Haar embedding. This paper isolates the exact mechanism. For operators  $A_c, A_f$  and a rectangular embedding  $J$ , set  $E = A_f J - J A_c$ . On the common resolvent set we prove

$$(z - A_f)^{-1} J - J (z - A_c)^{-1} = (z - A_f)^{-1} E (z - A_c)^{-1}.$$

Integration gives the corresponding Riesz-projector identity and a contour bound. For source states we further prove an exact two-term decomposition: channel transport equals a source defect passed through the fine projector plus the projector-transport defect acting on the coarse source. A dual formula controls observations and residues.

A 240-case complex audit has zero identity failures; maximum residuals are  $2.42 \times 10^{-16}$  for the resolvent formula and  $1.12 \times 10^{-14}$  for the channel decomposition. Applied to RH-202, the identity explains why a contour estimate cannot close without a new map: the quartet-restricted intertwining defects are 0.236–0.656, while the source defect is at least 0.781. The paper supplies an exact criterion, not a physical resolvent bound or an all-level transport theorem.

## 1 Rectangular interlevel problem

Let  $X_c = \mathbb{C}^{n_c}$  and  $X_f = \mathbb{C}^{n_f}$  with  $n_f \geq n_c$ . We do not assume that  $A_c$  is a compression of  $A_f$ . Let

$$A_c : X_c \rightarrow X_c, \quad A_f : X_f \rightarrow X_f, \quad J : X_c \rightarrow X_f. \quad (1)$$

The basic defect is

$$E = A_f J - J A_c. \quad (2)$$

When  $J$  is the Haar embedding,  $E$  measures both discretization change and the physical change in noise scale. This distinction matters: refinement alone does not imply  $E \rightarrow 0$ .

The standard perturbation theory of spectral subspaces is usually stated for two operators on one space [1, 2]. The rectangular identity below is the appropriate interlevel form.

## 2 Resolvent intertwining identity

Write  $R_f(z) = (zI - A_f)^{-1}$  and  $R_c(z) = (zI - A_c)^{-1}$ .

**Theorem 2.1** (Rectangular resolvent identity). *If  $z \in \rho(A_f) \cap \rho(A_c)$ , then*

$$R_f(z)J - JR_c(z) = R_f(z)ER_c(z). \quad (3)$$

*Proof.* Using  $R_f A_f = zR_f - I$  and  $A_c R_c = zR_c - I$ ,

$$\begin{aligned} R_f E R_c &= R_f (A_f J - J A_c) R_c \\ &= (zR_f - I) J R_c - R_f J (zR_c - I) \\ &= R_f J - J R_c. \end{aligned}$$

No equality of dimensions or invertibility of  $J$  is used. □

The formula separates geometry from spectral amplification. A moderate  $E$  can still produce a large projector defect when either resolvent is large, as is common for nonnormal operators [3].

## 3 Riesz projectors and a contour budget

Let  $\Gamma$  be a positively oriented rectifiable contour contained in both resolvent sets. Define

$$P_f = \frac{1}{2\pi i} \int_{\Gamma} R_f(z) dz, \quad P_c = \frac{1}{2\pi i} \int_{\Gamma} R_c(z) dz. \quad (4)$$

**Corollary 3.1** (Riesz transport). *The exact interlevel projector defect is*

$$P_f J - J P_c = \frac{1}{2\pi i} \int_{\Gamma} R_f(z) E R_c(z) dz. \quad (5)$$

Consequently,

$$\|P_f J - J P_c\| \leq \frac{|\Gamma|}{2\pi} \sup_{z \in \Gamma} \|R_f(z)\| \sup_{z \in \Gamma} \|R_c(z)\| \|E\|. \quad (6)$$

The same contour must isolate the intended clusters at both levels. A post hoc contour around two unrelated packets would make the formula true but would not establish branch continuity.

## 4 Source-channel decomposition

Matrix source states introduce a second embedding. Let  $S_c \in \mathbb{C}^{n_c \times m_c}$ ,  $S_f \in \mathbb{C}^{n_f \times m_f}$  and  $K : \mathbb{C}^{m_c} \rightarrow \mathbb{C}^{m_f}$ . Define

$$\Delta_S = S_f - J S_c K^*. \quad (7)$$

**Theorem 4.1** (Two-term channel budget). *For arbitrary projectors  $P_c, P_f$ ,*

$$P_f S_f - J P_c S_c K^* = P_f \Delta_S + (P_f J - J P_c) S_c K^*. \quad (8)$$

*If  $K$  is an isometry, then*

$$\|P_f S_f - J P_c S_c K^*\|_F \leq \|P_f\| \|\Delta_S\|_F + \|P_f J - J P_c\| \|S_c\|_F. \quad (9)$$

*Proof.* Add and subtract  $P_f JS_c K^*$  and apply submultiplicativity. No spectral assumption is needed for the algebraic identity.  $\square$

Thus a good projector map is insufficient when the physical source itself changes substantially. Conversely, a perfectly embedded source cannot repair a bad spectral projector map.

## 5 Observation and residue budget

Let  $O_c : \mathbb{C}^{n_c} \rightarrow \mathbb{C}^{m_c}$  and  $O_f : \mathbb{C}^{n_f} \rightarrow \mathbb{C}^{m_f}$ . Define

$$\Delta_O = O_f - KO_c J^*. \quad (10)$$

Applying Theorem 4.1 to adjoints gives a two-term formula for  $P_f^* O_f^* - JP_c^* O_c^* K^*$ .

More explicitly, with  $\Delta_{O^*} = O_f^* - JO_c^* K^*$ ,

$$P_f^* O_f^* - JP_c^* O_c^* K^* = P_f^* \Delta_{O^*} + (P_f^* J - JP_c^*) O_c^* K^*. \quad (11)$$

The transfer residue of a cluster is

$$r = \text{tr}(OPS). \quad (12)$$

After inserting and subtracting transported source, observation, and projector factors,  $r_f - r_c$  is bounded by three physical defects multiplied by the remaining factor norms. This makes clear that residue transport is strictly more demanding than eigenvalue matching.

There is also a direct channel-state bound. Put  $\widehat{X}_c = JX_c K^*$  and  $\widehat{Y}_c = JY_c K^*$ . Isometry of  $J, K$  preserves the coarse Frobenius pairing, and therefore

$$|r_f - r_c| \leq \|Y_f - \widehat{Y}_c\|_F \|X_f\|_F + \|\widehat{Y}_c\|_F \|X_f - \widehat{X}_c\|_F. \quad (13)$$

Combining (9), (11), and (13) gives a fully typed finite residue budget. It requires no eigenvector gauge, only projector, source, and observation estimates.

## 6 Identity audit

We generated 120 random complex rectangular operator pairs with dimensions 2–7, random isometric embeddings, and spectral parameters well outside both spectra. The maximum residual in (3) is

$$2.4157 \times 10^{-16}.$$

An independent 120-case audit used arbitrary projectors, sources, and row and column isometries in (8). Its maximum Frobenius identity residual is  $1.1178 \times 10^{-14}$ . All 240 cases pass the  $10^{-10}$  gate.

These checks validate implementation, not the physical assumptions of a common contour.

## 7 Application to the physical quartet

RH-202 gives the following inherited ranges:

| quantity                               | finite range or extremum |
|----------------------------------------|--------------------------|
| quartet-restricted relative $E$ defect | 0.23623–0.65573          |
| minimum relative source defect         | 0.78134                  |
| maximum oblique-projector defect       | 2.29068                  |
| maximum relative residue displacement  | 8.18374                  |

The right side of (6) cannot be small merely from these data. Moreover, the source term in (9) is already order one before resolvent amplification is considered.

This is the exact explanation of the RH-202 obstruction: the naive map fails in at least two independent places. Improving only the contour calculation would not make the source defect disappear.

## 8 Necessary conditions for a successful map

A future scale-dependent map  $J_\sigma$  should provide all of:

1. a common contour with controlled  $R_f$  and  $R_c$ ;
2. a small intertwining defect  $A_f J_\sigma - J_\sigma A_c$  on the packet;
3. compatible row and column source embeddings;
4. a dual observation estimate;
5. nonvanishing residues after transport.

These are sufficient ingredients for finite packet transport. Uniform versions would still be needed for Gate A.

## 9 Norm choice and validation strategy

Equation (6) is valid in every subordinate operator norm. Spectral norms are natural for principal-angle control; Frobenius norms are natural for matrix channel states. Mixing them is legitimate only with the explicit submultiplicative steps used in (9).

For a validated implementation, one should bound the resolvents directly on  $\Gamma$ , not infer them from distance to the spectrum. The latter is unsafe for nonnormal matrices. A quadrature approximation also needs a separate integration remainder or an argument-principle count. These requirements explain why the exact identity is a roadmap rather than a certificate by itself.

## 10 Claim boundary and next step

The resolvent, projector, and channel identities are exact finite theorems. The physical use is diagnostic only: no supremum resolvent bound or interval Riesz contour has been supplied. RH-204 next asks a weaker question that does not require state transport: whether the two conjugate spectral branches themselves possess a unique correspondence across levels and channels.

## References

- [1] Tosio Kato. *Perturbation Theory for Linear Operators*. Springer, 1995.
- [2] G. W. Stewart and Ji guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- [3] Lloyd N. Trefethen and Mark Embree. *Spectra and Pseudospectra*. Princeton University Press, 2005.